

When Banks Leak: Data Breaches and Loan Contract Design

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Showcase demo



- 1 Motivation
- 2 Hypotheses
- 3 Data & Design
- 4 Main Results
- 5 Mechanisms & Heterogeneity
- 6 Real Effects & Conclusion



Motivation



Why this matters

- Banks face cyberattacks far more often than other industries and account for $\sim 20\%$ of reported cyber incidents (2004–2023).
- Banks are not just capital providers — they *process information, monitor borrowers, and certify credit* (Diamond 1984; Fama 1985).
- A data breach is an **operational failure** that damages **reputational capital** while leaving core intermediation largely intact.

Setting: the syndicated loan market

Multidimensional, publicly observed contracts (price, size, maturity, collateral, covenants, syndicate) — ideal to detect how a bank *recalibrates* lending after a reputational shock (Sufi 2007; Schwert 2018).



Question

How do banks adjust syndicated loan contract design following their own cybersecurity data breaches?

- ① How banks **operationally respond** to breaches — coordinated changes across *multiple* contract dimensions.
- ② Contractual consequences of **cyber risk** for bank–borrower relationships.
- ③ **Reputational capital** in financial markets — a clean shock that hits perceptions but not capabilities; we quantify its contractual cost.



Hypotheses



Two competing predictions

H1 — Weakened bargaining power

Breach $\Rightarrow$ reputational damage $\Rightarrow$ borrowers gain leverage (information-leakage risk, signalling a compromised counterparty) $\Rightarrow$ bank **concedes** more favorable terms to retain relationships.

H2 — Conservative lending

Breach reveals control weaknesses / strains the bank $\Rightarrow$ bank **tightens**: higher rates, less credit, more restrictive terms.

Net effect is an empirical question

Adjustments may be *asymmetric* across products: defend relationship-intensive syndicated lending while retrenching in commoditized segments.



Data & Design



- **Breaches:** Privacy Rights Clearinghouse, 2005–2017. 9,014 events → exclude govt/nonprofit/edu → 1,374 Compustat firms → 420 financial → **109 bank breaches** (lead-arranger banks).
- **Loans:** DealScan syndicated facilities; Compustat via Chava–Roberts link; contract dated 90 days before facility start.
- **Final sample:** **14,497 loans**, **94** lead banks, **2,690** U.S. borrowers.

Breach heterogeneity (cyberattacks, insider theft, lost tapes...) adds noise ⇒ estimates are **lower bounds**; the same variation powers the severity tests.



$$\text{LoanTerm}_{i,t,l,b} = \alpha + \beta_1 \#Breaches_b + \mathbf{X}_{i,t-1} + \mathbf{L}_l + \mathbf{B}_{b,t-1} + \text{FE} + \varepsilon$$

- $\#Breaches$ = breaches at the lead arranger in the 360 days before the contract. DVs: **Log(Spread over LIBOR)**, **Log(Amount)**.

Three identification strategies

- 1 **Stacked DiD** for staggered timing (Cengiz et al. 2019; Baker et al. 2022).
- 2 **Propensity-score matching** of breached vs. similar non-breached banks.
- 3 **Within-borrower**: same borrower, same year, breached vs. non-breached lender.



Main Results



Main result: breached banks concede

Headline

The average breached bank **forgoes** $\approx$ **\$16 million in annual interest income per breach.**

After a breach, banks offer:

Larger loans	(higher Log Amount)
Lower spreads	(lower Log Spread over LIBOR)
Longer maturities	weaker lender protection
Less collateral	weaker lender protection
More shared lead roles	reputational co-insurance

- Consistent with **weakened bargaining power** (H1), not conservative lending (H2).
- Robust across stacked DiD, PSM, and within-borrower estimation.



Not a shift toward safer borrowers

- Breached banks do **not** expand total lending (but originate more corporate loans).
- Deposits: only marginally outflow; bank **profitability significantly declines**.
- **No systematic change in borrower risk profiles.**

Interpretation

Margins compress without depositor flight — retail depositors are insured; corporate borrowers face real leakage / reputational costs from a compromised counterparty. Reputational damage — not a safer-client shift — drives the contract changes.



Mechanisms & Heterogeneity



The reputational mechanism

Three cross-sectional tests, each sharpening the reputation story:

- ① **Breach severity** — records compromised and the stock-market reaction: more severe breaches $\Rightarrow$ more favorable borrower terms.
- ② **Borrower information risk** — high R&D intensity and high industry competition (HHI): more to lose from leakage $\Rightarrow$ larger concessions.
- ③ **Bank type** — effects concentrate in **commercial banks** (certification capital), *not* investment / smaller banks.

Why this rules out the fragility story

If borrowers feared lender *instability*, effects would cluster in smaller, weaker banks. They cluster in large commercial banks — **reputation, not fragility**.



Borrower heterogeneity & non-price terms

Bargaining power varies

- Outside options (CP eligibility, # past lenders) $\Rightarrow$ *more* favorable terms.
- Borrower's *own* breach history $\Rightarrow$ *less* favorable (bilateral reputation).
- Strong bank–borrower relationship *insulates* terms.

Non-price & syndicate

- Maturity $\uparrow$, collateral $\downarrow$.
- Covenant intensity: *no* change (standardized).
- Breached lead arrangers form *more co-arranging* partnerships; # participants unchanged.

Banks seek reputational *co-insurance* but cannot fully offset the deteriorated bargaining position.



Real Effects & Conclusion



Real effects on borrower investment

Borrowers receiving loans from breached banks — on favorable terms — respond:

- **Capital expenditure** ↑
- **Acquisition activity** ↑
- **R&D spending** — *no* effect (banks stay vigilant on high-risk intangibles even when bargaining power weakens)

Propagation

Cyber disruptions at financial intermediaries travel *beyond contract terms* into real resource allocation in the economy.



Conclusion

- A bank's own breach **erodes reputational capital** and weakens its bargaining power in syndicated lending.
- Banks concede across *many* margins (price, size, maturity, collateral, syndicate) — $\approx$ **\$16M** forgone interest per breach.
- Reputational capital — absent from balance sheets — carries **tangible contractual and real consequences**.
- Informs regulators on systemic cyber risk and managers on cybersecurity investment.

Slides generated by academic-latex-slides from the SSRN working paper, as a showcase example.



Appendix — Backup Slides



- PRC breach events 2005–2019: **9,014**.
- — government / nonprofit / education: **–1,921**.
- Matched to Compustat (parent-consolidated): **1,374**.
- Financial institutions (SIC 6000–6999): **420**; exclude nonbanks $\Rightarrow$ **109 bank breaches**.
- Lead arrangers hand-matched: 474 names $\rightarrow$ 113 gvkeys; final **94** distinct lead banks, **14,497** facilities, **2,690** borrowers (2005–2017).



$$LoanTerm_{i,t,l,b} = \alpha + \beta_1 \#Breaches_b + \mathbf{X}_{i,t-1} + \mathbf{L}_l + \mathbf{B}_{b,t-1} + \text{FE} + \varepsilon_{i,t,l,b}$$

- Treatment: # breaches at lead arranger(s) within 360 days before contract date (contract = facility start – 90 days).
- Borrower **X**: Log(Asset), Leverage, Tobin's q, Profitability, Z-score, rating indicators.
- Loan **L**: Log(Amount/Maturity), security, covenants, type.
- Bank **B** + borrower / year / loan-type / lender FE.



A3 — Three identification designs

- 1 **Stacked DiD** — event-specific cohorts avoid bad forbidden-comparison bias under staggered timing (Cengiz et al. 2019; Baker et al. 2022).
- 2 **PSM** — breached banks paired with observationally similar non-breached banks.
- 3 **Within-borrower** — same borrower, same year, breached vs. non-breached lender; absorbs borrower-year demand.

Inferences robust across all three; no evidence of a shift toward systematically safer borrowers.



- **Severity:** records compromised; breach-window stock return.
- **Borrower info risk:** R&D intensity; industry HHI.
- **Bank reputation:** commercial vs. investment bank.
- **Outside options:** commercial-paper eligibility; # of past lenders.
- **Bilateral reputation:** borrower's own breach history.
- **Relationship:** share of borrowing from current lender over 5 years (Bharath et al. 2011).

